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Temporal Transcriptome Analysis Suggests Modulation of Key Pathways and Hub Genes in a Mice Model of Multi-Drug Resistant (MDR) *Pseudomonas aeruginosa* Endophthalmitis

Poonam Naik[a,b] and Joveeta Joseph[a]

[a]Jhaveri Microbiology Centre, Brien Holden Eye Research Centre, L. V. Prasad Eye Institute, Hyderabad, Telangana, India; [b]Manipal Academy of Higher Education, Manipal, Karnataka, India

ABSTRACT

Purpose: Increasing incidence of multidrug-resistant *Pseudomonas aeruginosa* (*MDR-PA*) causing endophthalmitis challenges our ability to manage this vision-threatening condition. In this study, temporal dynamics of immune response in a mouse model of *MDR-PA* endophthalmitis was investigated by whole transcriptome analysis.
Methods: C57BL/6 mice were infected with *MDR-PA* and antibiotic susceptible *(S-PA)* clinical strains and disease severity were monitored at 6 and 24-h postinfection (p.i), following which eyeballs were enucleated. Microarray analysis was performed using SuperPrint G3 Mouse Gene Expression v2 chip and the differential gene expression analysis was performed with limma package in R (v4.0.0.)/Bioconductor (v3.11).
Results: Histopathological analysis revealed a significant difference in retinal architecture and vitreous infiltrates at 6 and 24 h. In comparison to *S-PA*, *MDR-PA* revealed altered expression of 923 genes at 6 h and 2220 genes at 24 h. Further, 23 and 76% of these altered genes and its downstream interacting proteins showed time-specific expression (6 and 24 h respectively), indicating their association with disease progression. At 24 hours, *MDR-PA* induced endophthalmitis showed aberrant immune response with the enrichment inflammasome signalling, dysregulated ubiquitination, complement cascade, MMPs NF-$\kappa\beta$ and IL-1 signalling.
Conclusion: The rapid development of transcriptional differences between the two-time points reveals that distinct genes contribute to disease severity. The results from this study highlighted a link between innate and adaptive immune responses and provided novel insights in the pathogenesis of *MDR-PA* endophthalmitis by extending the number of molecular determinants and functional pathways that underpin host-associated damage.

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Pseudomonas aeruginosa; endophthalmitis; multi-drug resistance; transcriptomics; murine model

Introduction

Endophthalmitis is defined as an inflammation of the intraocular structures, resulting from colonization of infectious agents like bacteria in the intraocular cavity. While an increase in drug-resistant pathogens in endophthalmitis was observed worldwide, we found *Pseudomonas* to be the most common isolate in 72.73% of multidrug-resistant (MDR) bacterial endophthalmitis cases.[1] In a clinical setting, *MDR-Pseudomonas aeruginosa (MDR-PA)* endophthalmitis is often associated with fulminant infection and poor visual prognosis even with early treatment with second line of intravitreal antibiotics thereby leading to a higher rate of evisceration and enucleation.[2–5] A study by Bawankar et al, (2019), reported that despite early vitrectomy and prompt administration of intravitreal antibiotics, the outcome was poor in about 50% of patients and 46% of patients underwent evisceration or phthisis.[5] Another study by Parchand et al (2020), revealed that the overall outcome was poor in *MDR-PA* induced endophthalmitis patients (85.5%), of which 37% of patients needed evisceration and 30.6% of the eyes developed phthisis bulbi while only 14.5% of the patient had >20/200 visual acuity, despite the usage of topical steroids.[6] Studies have shown that best results are obtained when the therapy is instituted early in the course of infection, preferably during the first 36 hours of the onset of symptoms, and when the infective organism is of low virulence.[7,8] Time being an important factor for optimum visual outcome, it is essential to understand the molecular basis of temporal host immune response affecting the pathogenesis of the disease.[9] In the past few years, extensive studies from our laboratory have provided considerable evidence on time-dependent accelerated immune response in human microglia and retinal pigment epithelial cells by *MDR-PA* infection. We observed a time-dependent upregulation of IL-1β, IL-6, TNF-α, IFN-γ, IL-10, IL-8, MMP-9, and GM-CSF in *MDR-PA* endophthalmitis compared to antibiotic susceptible strains (*S-PA*).[10,11] A study by Rajamani et al., (2016) deciphered the temporal immune response in *Staphylococcus*

CONTACT Joveeta Joseph joveeta@lvpei.org Jhaveri Microbiology Centre, Brien Holden Eye Research Centre, L. V. Prasad Eye Institute, Hyderabad, Telangana 500034, India

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aureus endophthalmitis which provided deep insight into the genes, pathways, and biological networks that are perturbed during different phases of infection.[12] Our earlier study in a murine model of *Pseudomonas aeruginosa* endophthalmitis reported a shift toward a more immunogenic profile including an increase in inflammatory response, retinal dissipation, and aberrant pathway regulation at an earlier time point than the *S-PA* strain found highly dysregulated pathways like chemokine receptor binding, TNF signalling, Toll-like receptor and NOD-like receptor signalling, NF-κB signalling, and MAP kinase signalling in *MDR-PA*-infected eyes.[13] In this study, we observed that there is a time-dependent disease progression as evident by higher retinal degeneration and inflammation, and the expression of many differentially regulated genes which were specific to the later time point (24 h) only. The above observation led us to hypothesize there might be a correlation between the differential expression of genes and the corresponding time point leading to higher disease severity. However, to the best of our knowledge, there is no report on temporal host immune response correlating with the disease severity in *MDR-PA* endophthalmitis. Thus, considering the knowledge gap, in this study, we performed a comparative transcriptomic analysis of *MDR-PA* infected ocular tissue at an early (6 hours) and later (24 hours) stage of infection, compared to *S-PA* in mice, and highlighted the regulations of host pathways. Additionally, an overtime observation of gene expression allowed us to find fine variations in the kinetics of immune pathways activation. The results presented in this study provided a detailed understanding of the host gene expression profile and represent an essential step for identifying novel factors involved in the process of disease progression and increased retinal degeneration and could provide insight into the signature molecules for the rational design of therapeutic targets.

Methods

Ethics statement

The study was approved by the Institutional Review Board of the L. V. Prasad Eye Institute (LEC 09-18-125). The procedures related to mice were conducted according to guidelines and recommendations of the Guide for the Care and Use of Laboratory Animals by Institutional animal ethics committee (IAEC), Sipra Labs, Hyderabad, India. They approved the study under protocol SLL-PCT-IAEC/20-21/B dated 08/20/2020. The animals were also maintained according to institutional guidelines and the ARVO Statement for the Use of Animals in Ophthalmic and Vision Research.

Antibiotic susceptibility testing and bacterial culture preparation

The clinical strains selected for the study were isolated from the vitreous of patients diagnosed with culture-proven endophthalmitis seen at LV Prasad Eye Institute, Hyderabad with identification numbers as L-2050/18 (*S-PA*) and L-2051/18 (*MDR-PA*). The Antibiotic susceptibility profiles of all the strains were checked using conventional microbiologic protocols using ViTEK-2 (Automated Microbial System; Biomerieux Vitek, Hazelwood, Missouri, USA) which provided "breakpoint" MICs (minimal inhibitory concentration values). This was further confirmed by E-test (Biomerieux). The two strains, namely, *S-PA* (L-2050/18), was susceptible to all routine antibiotics like Ciprofloxacin, gentamycin, ofloxacin, amikacin, moxifloxacin, gatifloxacin, ceftazidime, imipenem, piperacillin-and tazobactam, and *MDR-PA* (L-2051/18) was resistant to all the aforementioned antibiotics.

Experimental P. aeruginosa endophthalmitis model

The experiments described here involved the use of C57BL/6 mice ($\sim$ 8 weeks of age) and were performed at Sipra Labs, Hyderabad, which is an outsourcing animal facility. Endophthalmitis was induced in the eyes of C57BL/6 mice by intravitreal injection of 10,000 CFU of *S-PA* or *MDR-PA* as described previously.[13] The contralateral eyes were injected with sterile PBS as surgical controls. The PBS injected eyes were processed for histopathological comparison, and microarray analysis to obtain baseline values. The *S-PA* ($n = 3$) and *MDR-PA* ($n = 3$) infected eyeballs were subjected to histopathological analysis at both 6 h and 24 h. The microarray experiment involved PBS ($n = 3$), *S-PA* ($n = 3$), and *MDR-PA* ($n = 3$) injected eyes at the aforementioned time points. In total, we have used 24 mice for these experiments. All eyes were assessed throughout the course of infection by slit-lamp biomicroscopy, while histologic analysis, along with microarray analysis, were performed after enucleation of the eye at 6 and 24 h post-infection (p.i.).

Clinical evaluation

The clinical changes occurring during the induced experimental endophthalmitis were scored independently by an ophthalmologist who was blinded to the type and time of injection. The scoring was done using a handheld slit-lamp biomicroscope (PSLAIA-11, Appasamy associates, Chennai, India). Clinical scoring was assigned on a scale from 0 to 4+ based on the criteria described earlier.[14] The clinical severity was assessed based on the inflammation in the anterior segment, vitreous, presence/absence of red reflex, and retinal clarity. Clinical changes were graded Photographs of mouse eyes were taken for visualization of progression of disease severity.

Histopathological analysis

Eyeballs were enucleated and immediately immersion-fixed in 4% paraformaldehyde in 0.1 M PBS then embedded in paraffin at indicated time points and were processed for cryosectioning as previously described for histological analysis.[13] The sections were then stained with hematoxylin and eosin (H&E).

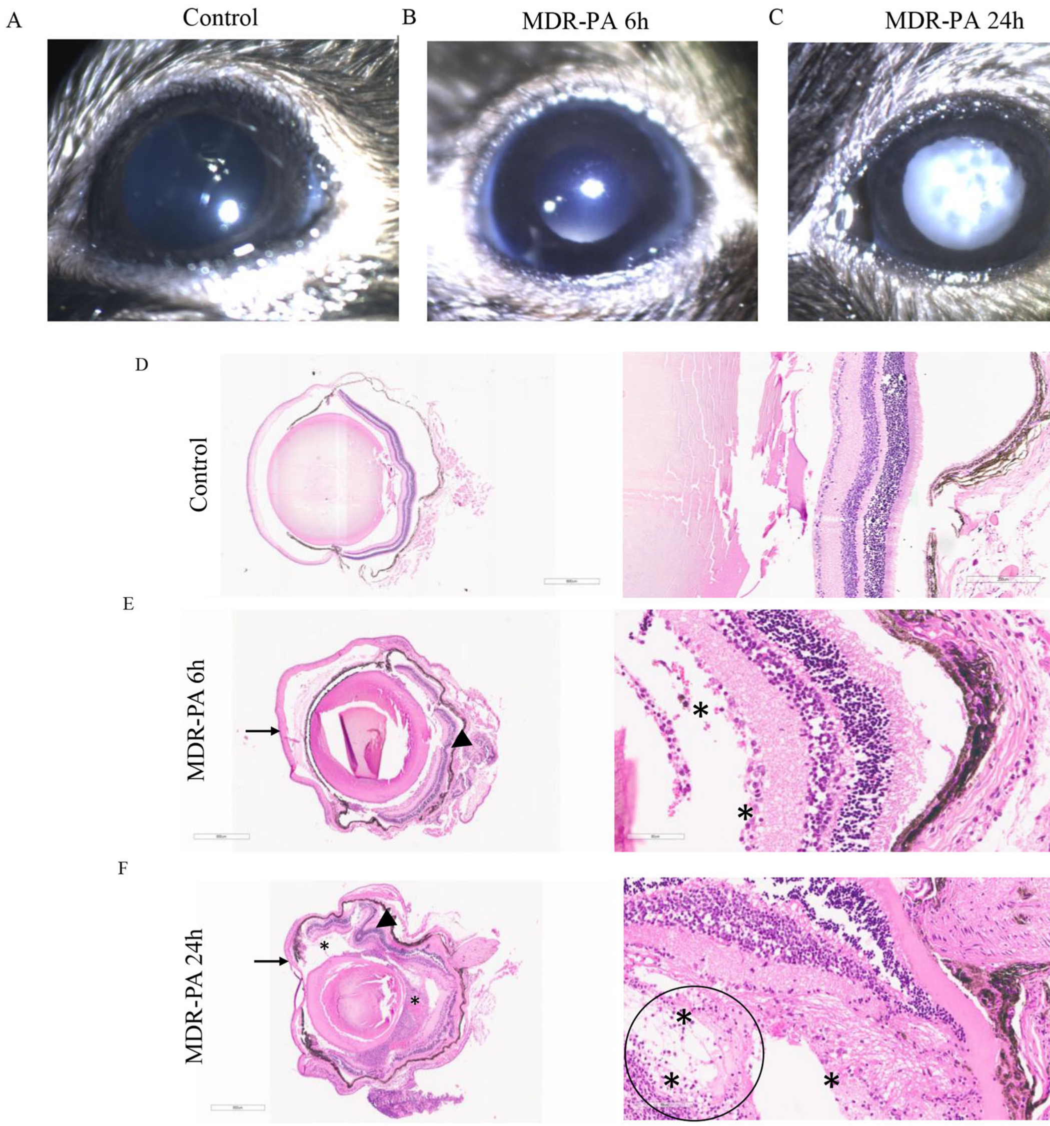

Figure 1. Representative photographs of the C57BL/6 mice eyes injected with PBS (A) 104 CFU of MDR-PA for 6 h (B) and 24 h (C). All eyes were assessed and harvested at 6 and 24 hours post-infection. Clinical score related to anterior segment inflammation, presence/absence of red reflex, vitreous inflammation, and retinal clarity at 6 and 24 h. Representative H&E-stained sections of whole eye, control (D); Original magnification 2.5× (left panel) and 13× (right panel), 6 h©); Original magnification 2.5× (left panel) and 35.5× (right panel) and 24 h (F); Original magnification 2.5× (left panel) and 37.6× (right panel) post-infection (p.i). Pathological features indicated: retinal folding (arrowhead), inflammatory infiltrates (asterisk), corneal edema (arrow), and circle denotes dense lymphocytic infiltration.

RNA sample preparation and microarray hybridization

Total RNA from mice eyeball ($n = 3$) from each group was isolated separately by Qiagen RNA extraction followed by DNase I treatment. The quality and quantity of isolated RNA were evaluated by determining RNA integrity number (RIN) using an in-house Agilent 2100 Bioanalyzer (Agilent Technologies, Santa Clara, CA) with nanochip systems. The transcriptomic profiling of mice retinal tissue was performed at two time points (6 h and 24 h) postinfection with *MDR-PA* and *S-PA*. To identify statistically significant differences in gene expression, the microarray analysis was performed on 3 separate biological replicates (3 separate eyeballs) at each time point using SurePrint G3 Mouse Gene Expression Microarrays (Agilent) containing 60,000 probes. The arrays were scanned using the in-house Agilent Scanner G2505C (Agilent Technologies, Inc.). Agilent Feature Extraction software (version 11.0.1.1; Agilent Technologies, Inc.) was used to analyze the acquired array images.

Transcript annotation and analysis

The probe sets of two dye arrays were annotated with Entrez gene identifiers using the Bioconductor. The differentially expressed gene (DEG) analysis was performed with limma package in R (v4.0.0.).Bonferroni correction was applied to p-value. Data were filtered using a fold change cut-off of 2.0 and p adj ≤ 0.1. The enrichment of the pathways was done using Reactome pathway database (Functional Enrichment analysis tool (FunRich V3.1.1)). Venn diagrams were created using Venny (1.0) to illustrate the unique data sets over the two time points. Interactive

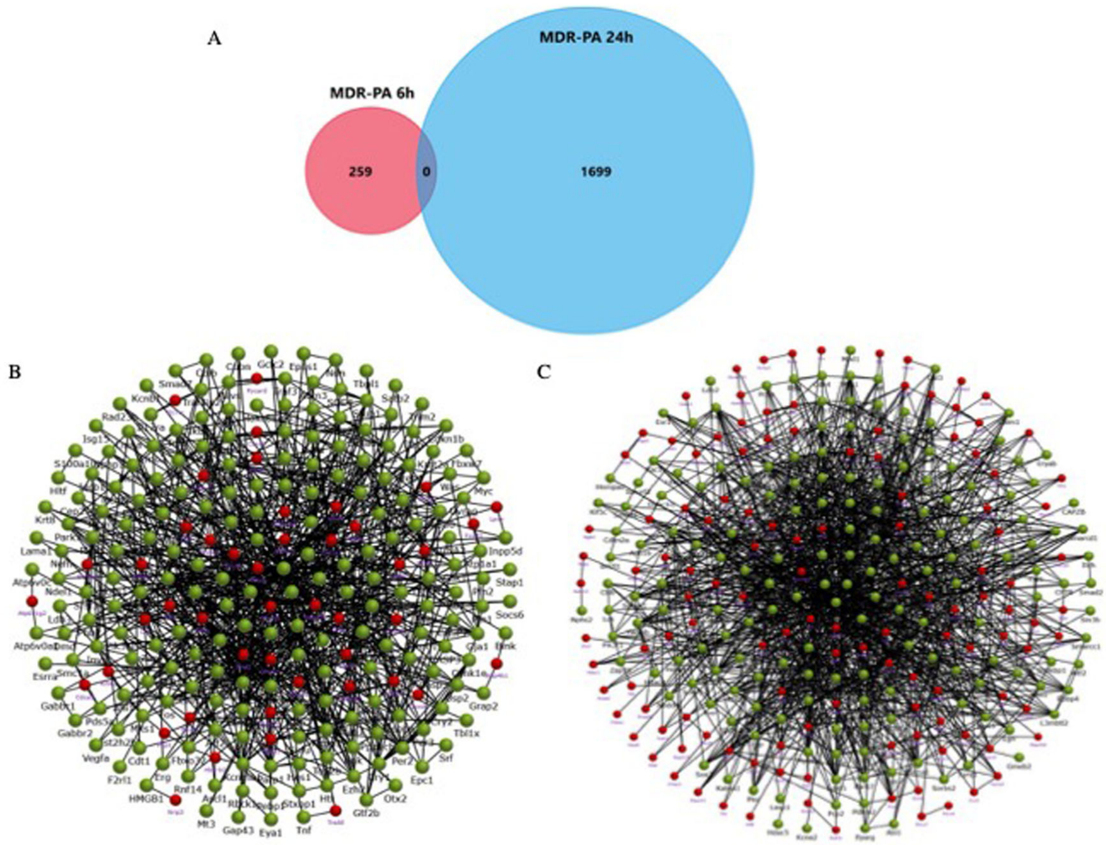

Figure 2. C57BL/6 mice eyes infected with S-PA and MDR-PA for 6 h ($n = 3$, each group) and 24 h ($n = 3$, each group) were subjected RNA extracted, amplified, labelled and used to hybridize Agilent microarrays. Genome-wide transcriptome expression profiling of *MDR-PA* infected mice eye was performed in comparison with S-PA infected mice eye. Venn diagrams showing differentially expressed genes in response to *MDR-PA* at 6 h and 24 h (A). PPI network analysis for time-specific genes revealed interactions with each other and with other proteins at both 6 h (B) and 24 h (C). 24 h timepoint had the highest degree in this network (network center). Statistical threshold of fold change ≥ 2.0 and p adj ≤ 0.1 were applied for both.

Network Analysis was performed using Funrich and string db (https://string-db.org).

Results

In vivo model of endophthalmitis

The clinical features of our mouse model of *P. aeruginosa* endophthalmitis resembled the clinical course of human endophthalmitis, such as anterior segment inflammation, diminution of red reflex, vitreous inflammation, and retinal clarity ($n \geq 12$ in each group). None of the control mice exhibited any signs of inflammation at any time postinjection though retinal detachment was observed as a result of histopathological processing (Figure 1(A)). In comparison mice at 6 h displayed mild inflammation with minimal vitreous infiltrate and slight retinal folding (Figure 1(B)), in contrast, the clinical signs were more evident at 24 h ($n = 3$) (Figure 1(C)). Pathological features included increased robust vitreous infiltrate, significant corneal edema and extensive retinal structural damage (e.g. increased inner and outer retinal thickness, retinal folds). Accumulation of inflammatory cells in the anterior chamber and vitreous cavity which also occurs in patients with endophthalmitis was greater at 24 h. Importantly, at 6 h not much of retinal damage was observed. Moreover, 24 h sections exhibited periocular inflammation.

Differntial gene expression

To provide a framework to understand how genes involved in the host immune response are regulated to respond to *MDR-PA*, we compared the mRNA population in a time-dependent manner. Transcriptomes of *MDR-PA*-6 h and MDR-PA-24 shared a similarity comprising 521 genes in overall gene expression. Although an overlap, transcriptomics exhibited a distinct and broader gene set at 24 h, suggesting a major shift in gene expression compared to 6 h. The data analysis was done in comparison to *S-PA* to obtain differentially expressed genes that would be specific to host-*MDR-PA* interactions. A total of 923 genes at 6 h and 2220 genes at 24 h were significantly differentially regulated *in MDR-PA* infected group (FC ≥ 2, adjusted *p-value* < 0.1) compared to the *S-PA* infected group. Interestingly, we observed that a total of 259 genes were found to be expressed exclusively at 6 h while 1699 genes were expressed at 24 h with no overlapping genes (Figure 2(A)). This suggested that *MDR-PA* infection initially triggers an immune response, manifesting as early qualitative and quantitative changes in gene transcription, which grows rapidly and spreads exponentially over time compared to *S-PA* infected mice eyes. Next, the time bound DEGs were subjected to functional interaction analysis and compared to 6 h (Figure 2(B)), we found a greater number of interacting nodes at 24 h (Figure 2(C)) time point using Funrich suggesting a

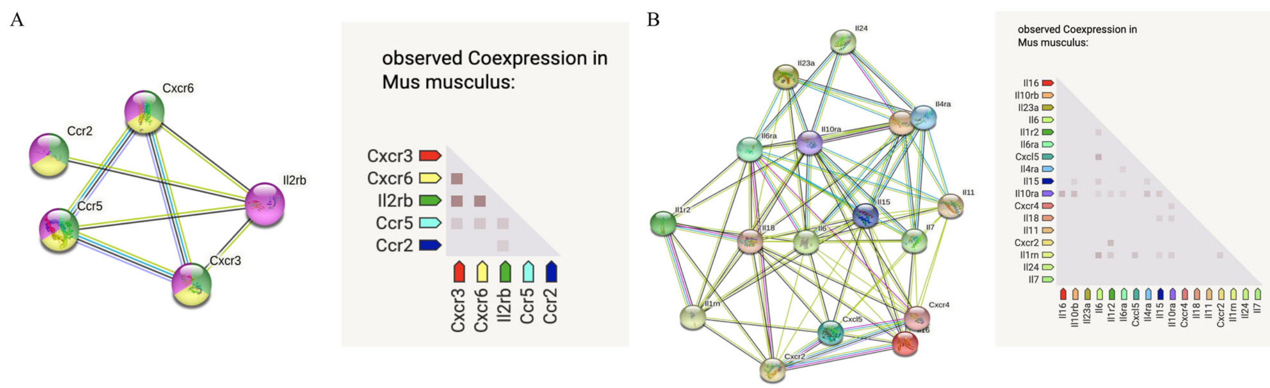

Figure 3. PPI network and co-expression analysis of the hub genes in *MDR-PA* endophthalmitis at 6 h(A) and 24 h (B) time points. The inflammatory mediators discussed were analyzed using the String database (version 11.0). The co-expression analysis of the hub genes using the STRING online database.

more perplexed host immune response at the later stage of infection.

The altered expression pattern with respect to time indicated distinguished functional categories involved in the disease pathogenesis, hence to follow transcriptional dynamics in a timewise manner, we looked into the functional aspect of the genes. Expression of 6 genes encoding adhesion proteins, like Desmocollin-3, Desmocollin-2, Integrin alpha-M, Protocadherin 10, Integrin beta-3, Cadherin-22 were upregulated during the early response. Adaptor molecules of apoptosis like TRADD was upregulated only at 6 h. Additionally, tight junctions proteins and collagens were upregulated at 6 h time point. In contrast, growth factors like fgf11, fgf9, and G-protein coupled receptor-associated sorting protein 2 (Gprasp2) and regulator of G protein signalling (Rgs 7) were found to be downregulated at an early phase of infection. At 24 h Lipocalin (LCN-2), Interferon, alpha-inducible protein 27 like protein 2 A (Ifi27la) which is suggested to be a novel regulator of neuroinflammation in microglia, LPS-induced TN factor (Litaf), CD-14 which is a marker for monocytes along with TRAF1 whose expression is largely limited to activated immune cells were expressed. GFAP which is a known marker of retinal stress was also found to be upregulated at 24 h.

Temporal differences in immune-related gene expression

Inflammatory mediators have been demonstrated to play master role in response to bacterial endophthalmitis. We observed that the early *MDR-PA* endophthalmitis response is characterized by a limited number of cytokines and chemokines like Interleukin IL-20rb, IL-2rb, C-X-C chemokine receptor CXCR-3, CXCR-6, CCR-2 and CCR-5. However, during the late hour (24 h) timepoint, a broader range of inflammatory mediators were found to be upregulated which includes IL-11, IL-15, IL-1receptor accessory protein, IL-4ra, IL-7r, IL-1r2, IL-10ra, IL-10rb, IL-6, TNF-α, IL-24, IL-16, IL-1α, IL-6ra, IL-23ra, IL-18bp, CCL-6, CCL-24, CCL-12, CCL-3, CCL-9, CCL-4, CCL-7, CCL-19, CXCL-2, CXCL-5, CXCL-12, CXCL-16, CXCL-13, CXCL-14, CXCL-9, chemokine like factor, CXCR-2 and CXCR-4. Additionally, the STRING online database was used to construct the PPI network of the inflammatory mediators expressed at 6 h (Figure 3(A)) and 24 h (Figure 3(B)). The results of the gene co-expression analysis of the 5 and 17 inflammatory mediators showed potential active interaction with each other.

In addition, we observed that components of the complement system which is a major part of innate immunity were also significantly overexpressed only at 24 h p.i. These include Cfb, C5ar1, C1qa, C1qb, C1qc, C3, C4b, while C1ql3, which stimulates glucose uptake in various cells, was found to be downregulated. Altogether this suggests an explosive immune response at the later time point, which is likely to be responsible for collateral retinal damage and a good percentage of vision loss.

Dysregulated ubiquitin regulators, inflammasome complex and MMPs at the later time point of infection

It is well known that protein ubiquitination plays a critical role in the initiation and termination of innate response pathways. Transcripts of ubiquitin D, Usp54, Usp53 were upregulated at 24 h p.i. At the other extreme, Usp33, Usp2 were downregulated at 24 h. Reduced USP33 has been shown to decrease the stability of suppressor molecules of major pro-inflammatory pathways.[15] This indicates that regulatory molecules related to proteasomal pathway are possibly involved in the pathogenesis of drug resistant endophthalmitis. We next explored, the expressional variation of matrix metalloproteinases (MMPs) in the *MDR-PA*. These included upregulation of MMP-25, MMP-8, MMP-3, MMP-9, and MMP-19, while the inhibitory molecules like TIMP-2 and TIMP-3 were downregulated at 24 h p.i. Additionally, Caspase-1, Caspase-4, Caspase-8, and NLRP3 were also found to be upregulated only at 24-h time point.

Immune pathways strongly associated with the severity of endophthalmitis

The extended changes in transcriptome markers at a late time point directed our attention to deciphering the underlying signalling pathway involved. Although the GO analysis showed overlap between the few enriched biological processes, the enriched Reactome pathways based on the Reactome database at 6 and 24 h were distinct. At 6 h p.i

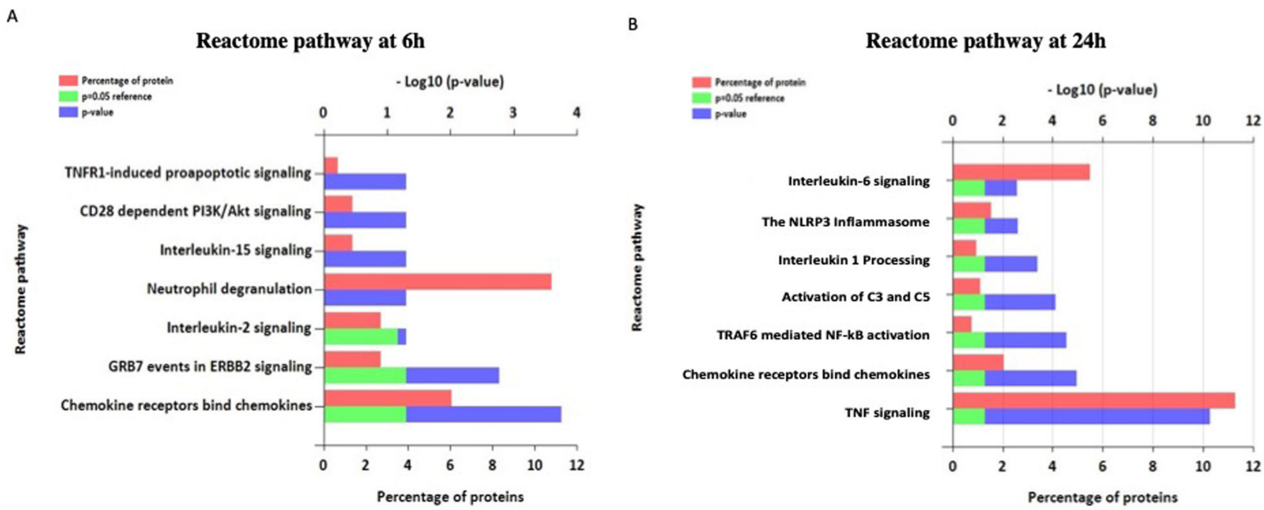

Figure 4. Functional gene enrichment analysis of all identified transcripts. Reactome pathways enriched in genes are more than 2-fold abundant compared to the *S-PA* group at 6 h (A) and 24 h (B). Bars represent the percentage of protein genes assigned to the indicated term, reference *p-value* (0.05), and the calculated *p-value* of enrichment for the indicated term.

IL-15, IL-2 signalling were among the top enriched pathways (Figure 4(A)). The downregulated proteins at 6 h, resulted in enrichment of FGF, G-protein receptor signalling. At the later time point, pathways like NF-κB, TNF, Inflammasome, and complement cascade were among the highly enriched ones (Figure 4(B)). The downregulated genes at 24 h enriched into TGF-β, PDGF signalling, Mannose metabolism, Glutamate receptor pathway, Glycolysis and Fructose galactose metabolism.

Discussion

The primary aim of this study was to identify genes that are temporally expressed during *MDR-PA* infection and understand their role in the disease severity. In our study, transcriptome analysis revealed that, at 24-h time point, genes related to ubiquitination were found to be dysregulated. There are reports which show that bacterial pathogens target ubiquitination to subvert and manipulate host defence systems.[15] These strategies enable bacterial pathogens to proliferate[16] and evade the host immune response,[17] although the role of *MDR-PA* in disrupting ubiquitin pathways has not been reported. Infiltration of inflammatory cells is mediated by adhesion molecules. Our study showed overexpression of multiple adhesion molecules at 6 h which correlates with a study by Giese et al (2000).[18] They showed that expression of adhesion molecule peaked at early time points in *S. aureus*–injected eyes, and returned to constitutive levels at 72 h. Furthermore, we also observed the expression of an increased number of inflammatory mediators at 24 h. Increased cytokine and chemokine responses have already been correlated with increased disease severity.[19,20] A study by Miller et al. suggested that host immune mediators are important components and the timing and level of their production may be related to the severity of the damage and visual outcome.[9] Taken collectively, our data suggest that the increase in inflammatory mediator transcripts may be explained by increased severity followed by rapid retinal deterioration at 24 h. MMPs play an important role in injury repair during inflammation, however, the excessive expression of MMPs can also destroy the ECM and promote further inflammation.[21,22] Therefore, it is not surprising that the retinal dissipation is higher at 24 h. It is important to confine complement activation to pathogenic surfaces due to the destructive potential of complement activation to the host. Therefore, effectors of complement activation need to be tightly regulated to maintain the balance.[23] In our study, we observed that a lot of complement molecules are expressed at 24 h. We also observed that the magnitude of upregulation was stronger in the 24-h transcriptome data set than in the 6-h data. Overall, the study of *MDR-PA* induced endophthalmitis within a time-dependent context is of crucial importance, as it is expected to shed light toward a better understanding of the progression of disease and thus, to improve clinical outcome. The concepts and data analysis discussed herein may be useful in objectively identifying coherent patterns and sequence of events in the complex signalling network of host response as well as developing targeted immunomodulation as a therapeutic approach in MDR endophthalmitis. While we identified dysregulated genes and pathways that are temporally expressed during MDR-PA infection, we could not study the impact of antibiotics on the altered genes and pathways. Treating the animals with antibiotics would have helped in investigating the disease severity compared to untreated ones. This would have added further insights and strengthened our findings. This remains a limitation in our study and our future studies would focus on the same. Further, only one strain of MDR-PA was used to investigate and characterize the differential immune response. Usage of additional strains would have further strengthened our conclusion and results. Overall, these results suggest that "early" responses to *MDR-PA* are relatively well controlled while the late response is characterized by aberrant immune response, inflammasome signalling and dysregulated ubiquitination. Our study demonstrates that the disease severity is linked to the host immune response and also reinforces the argument, that the host immune response plays an equivalent role in disease progression in MDR endophthalmitis.

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Ethical approval

The animal study was reviewed and approved by the Institutional Animal Ethics Committee (IAEC), Sipra Labs, Hyderabad, India (SLL-PCT-IAEC/20-21/B) dated 08/20/2020.

Author contributions

Conceptualization: JJ and PN, Methodology: PN, Software: PN. Validation: PN and JJ. Resources: JJ. Writing—original draft preparation: PN. Writing—review and editing: JJ. Project administration: JJ. Funding acquisition: JJ.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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ORCID

Joveeta Joseph http://orcid.org/0000-0002-8421-1977

Data availability statement

Raw data were generated at Prof. Brien Holden Eye Research Centre, LV Prasad Eye Institute. Derived data supporting the findings of this study are available from the corresponding author Dr. Joveeta on request.

References

1. Pathengay A, Moreker MR, Puthussery R, Ambatipudi S, Jalali S, Majji AB, Mathai A, Husssain N, Dave V, Sharma S, et al. Clinical and microbiologic review of culture-proven endophthalmitis caused by multidrug-resistant bacteria in patients seen at a tertiary eye care center in southern India. Retina. 2011;31(9):1806–1811.
2. Eifrig CW, Scott IU, Flynn HW, Jr., Miller D. Endophthalmitis caused by *Pseudomonas aeruginosa*. Ophthalmology. 2003;110(9):1714–1717. doi:10.1016/S0161-6420(03)00572-4.
3. Sridhar J, Kuriyan AE, Flynn HW, Jr, Miller D. Endophthalmitis caused by pseudomonas aeruginosa: clinical features, antibiotic susceptibilities, and treatment outcomes. Retina. 2015;35(6):1101–1106. doi:10.1097/IAE.0000000000000469.
4. Falavarjani KG, Alemzadeh SA, Habibi A, Hadavandkhani A, Askari S, Pourhabibi A. *Pseudomonas aeruginosa* endophthalmitis: clinical outcomes and antibiotic susceptibilities. Ocul Immunol Inflamm. 2017;25(3):377–381. doi:10.3109/09273948.2015.1132740.
5. Bawankar P, Bhattacharjee H, Barman M, Soibam R, Deka H, Chandra Kuri G, Medhi J. Outbreak of multidrug-resistant *Pseudomonas Aeruginosa* endophthalmitis due to contaminated trypan blue solution. J Ophthalmic Vis Res. 2019;14(3):257–266. doi:10.18502/jovr.v14i3.4781.
6. Parchand SM, Agrawal D, Chatterjee S, Gangwe A, Mishra M, Agrawal D. Post-cataract surgery cluster endophthalmitis due to multidrug-resistant *Pseudomonas aeruginosa*: A retrospective cohort study of six clusters. Indian J Ophthalmol. 2020;68(7):1424–1431. doi:10.4103/ijo.IJO_1612_19.
7. Peyman GA, Raichand M, Bennett TO. Management of endophthalmitis with pars plana vitrectomy. Br J Ophthalmol. 1980;64(7):472–475. doi:10.1136/bjo.64.7.472.
8. Kernt M, Kampik A. Endophthalmitis: pathogenesis, clinical presentation, management, and perspectives. Clin Ophthalmol. 2010;4(4):121–135. doi:10.2147/opth.s6461.
9. Miller FC, Coburn PS, Huzzatul MM, LaGrow AL, Livingston E, Callegan MC. Targets of immunomodulation in bacterial endophthalmitis. Prog Retin Eye Res. 2019;73:100763. doi:10.1016/j.preteyeres.2019.05.004.
10. Naik P, Singh S, Vishwakarma S, Kaur I, Dave VP, Kumar A, Joseph J. Multidrug-resistant *pseudomonas aeruginosa* evokes differential inflammatory responses in human microglial and retinal pigment epithelial cells. Microorganisms. 2020;8(5):735. doi:10.3390/microorganisms8050735.
11. Naik P, Singh S, Rudraprasad D, Dave VP, Kumar A, Joseph J. Multidrug-resistant *Pseudomonas aeruginosa* triggers differential inflammatory response in patients with endophthalmitis. Trans Vis Sci Tech. 2021;10(9):26. doi:10.1167/tvst.10.9.26.
12. Rajamani D, Singh PK, Rottmann BG, Singh N, Bhasin MK, Kumar A. Temporal retinal transcriptome and systems biology analysis identifies key pathways and hub genes in *Staphylococcus aureus* endophthalmitis. Sci Rep. 2016;6:21502. doi:10.1038/srep21502.
13. Naik P, Pandey S, Naik MN, Mishra DK, Boyenpally SR, Joseph J. Transcriptomic and histological analysis of exacerbated immune response in multidrug-resistant *pseudomonas aeruginosa* in a murine model of endophthalmitis. Front Immunol. 2022;12:789023.
14. Ramadan RT, Ramirez R, Novosad BD, Callegan MC. Acute inflammation and loss of retinal architecture and function during experimental *Bacillus* endophthalmitis. Curr Eye Res. 2006;31(11):955–965. doi:10.1080/02713680600976925.
15. Mishra R, Lahon A, Banerjea AC. Dengue virus degrades USP33-ATF3 axis via extracellular vesicles to activate human microglial cells. J Immunol. 2020;205(7):1787–1798. doi:10.4049/jimmunol.2000411.
16. Ashida H, Kim M, Sasakawa C. Exploitation of the host ubiquitin system by human bacterial pathogens. Nat Rev Microbiol. 2014;12(6):399–413. doi:10.1038/nrmicro3259.
17. Jiang X, Chen ZJ. The role of ubiquitylation in immune defence and pathogen evasion. Nat Rev Immunol. 2011;12(1):35–48. doi:10.1038/nri3111.
18. Giese MJ, Shum DC, Rayner SA, Mondino BJ, Berliner JA. Adhesion molecule expression in a rat model of *Staphylococcus aureus* endophthalmitis. Invest Ophthalmol Vis Sci. 2000; 41(1):145–153.
19. Deshmukh D, Chakrabarti M, Jayasudha R, Hasnat Ali M, Tyagi M, Sharma S, Joseph J. Elevated cytokine levels in vitreous as biomarkers of disease severity in infectious endophthalmitis. PLOS One. 2018;13(10):e0205292. doi:10.1371/journal.pone.0205292.
20. Giese MJ, Sumner HL, Berliner JA, Mondino BJ. Cytokine expression in a rat model of *Staphylococcus aureus* endophthalmitis. Invest Ophthalmol Vis Sci. 1998;39(13):2785–2790.

21. Chen M, Xu H. Parainflammation, chronic inflammation, and age-related macular degeneration. J Leukoc Biol. 2015;98(5): 713–725. doi:10.1189/jlb.3RI0615-239R.
22. Kumar A, Kumar A. Role of Staphylococcus aureus virulence factors in inducing inflammation and vascular permeability in a mouse model of bacterial endophthalmitis. PloS One. 2015;10(6): e0128423. doi:10.1371/journal.pone.0128423.
23. Merle NS, Church SE, Fremeaux-Bacchi V, Roumenina LT. Complement system part ©–- molecular mechanisms of activation and regulation. Front Immnol. 2015; 2(6):262.